



MANGAYARKARASI COLLEGE OF ENGINEERING

Department Of Computer Science Engineering

Completed a project on

CALCULATING FAMILY EXPENSES USING SERVICE NOW

Submitted By

Team ID : NM2025TMID01168

- 1. Team Leader : Amirtha Varshni R**
- 2. Team Member : Hamsavarthiny P**
- 3. Team Member : Lakshmi Priya A**
- 4. Team Member : Varsha V**

CALCULATING FAMILY EXPENSES USING SERVICE NOW

Problem Statement:

The project aims to develop a comprehensive expense calculation system using ServiceNow. This system will enable users to track and manage family expenses efficiently. It will include features such as expense categorization, budget setting, real-time tracking, and reporting capabilities. Utilizing ServiceNow's robust platform, the project will ensure seamless integration, user-friendly interface, and scalability to accommodate varying family sizes and financial complexities. The end goal is to empower users with the tools they need to make informed financial decisions and promote financial well-being within the family unit.

Objectives:

The objective of this project is to develop a family expense management system on the ServiceNow platform that enables efficient tracking, categorization, and budgeting of expenses. It aims to provide real-time monitoring, user-friendly dashboards, and insightful reports to support informed financial decisions. The system will ensure scalability for different family sizes, promote financial discipline through spending analysis, and maintain data security using ServiceNow's robust features.

Skills:

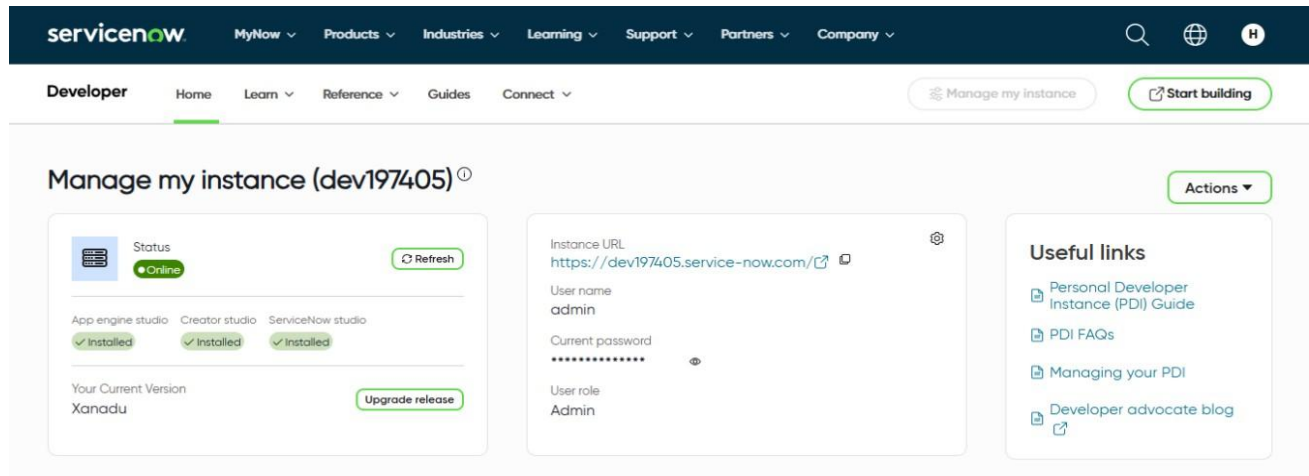
TASK INITIATION

Milestone 1 : Setting up ServiceNow Instance

Activity : Setting up ServiceNow Instance

1. Sign up for a developer account on the ServiceNow Developer site "<https://developer.servicenow.com>".
2. Once logged in, navigate to the "Personal Developer Instance" section.
3. Click on "Request Instance" to create a new ServiceNow instance.
4. Fill out the required information and submit the request.

5. You'll receive an email with the instance details once it's ready.
6. Log in to your ServiceNow instance using the provided credentials.

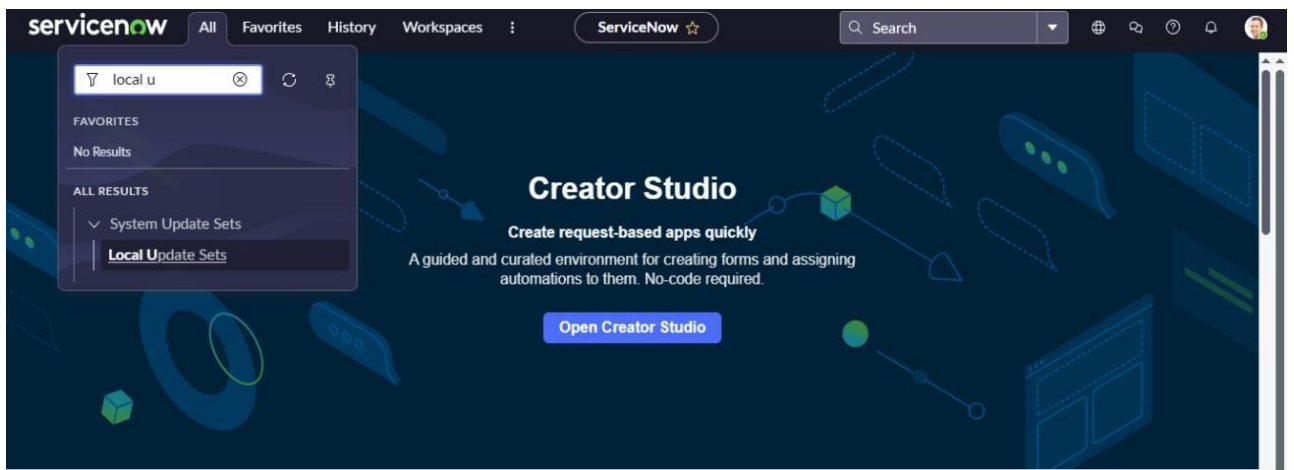


7. Now you will navigate to the ServiceNow.

Milestone 2 : Creation of New Update Set

Activity: Creation of New Update Set

1. Go to All >> In the filter search for Local Update set > click on New



2. Enter the Details as:
Name : Family Expenses
3. Then click on Submit and Make current

servicenow All Favorites History : Update Set - Create New Update Set ☆ Search

Update Set New record

* Name Application

State

Parent

Release date

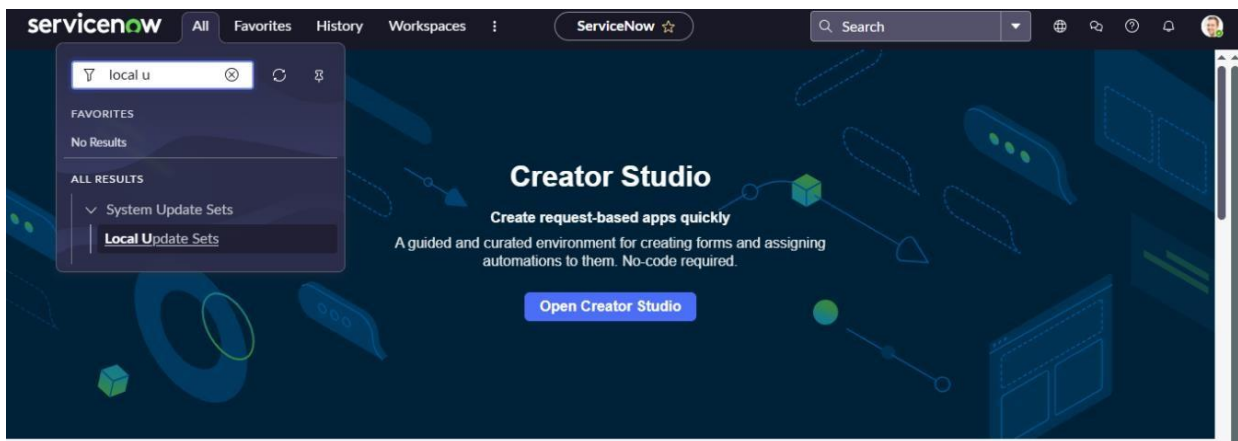
Description

Submit Submit and Make Current

Milestone 3: Creation of Table

Activity: Creation of New Update Set

1. Go to All >> In the filter search for Local Update set > click on New



2. Enter the Details as:
Name : Family Expenses
3. Then click on Submit and Make current.

servicenow All Favorites History Update Set - Create New Update Set Search

Update Set New record

* Name Family Expenses Application Global

State In progress

Parent

Release date

Description

Submit Submit and Make Current

Milestone 4: Creation of Table

Activity 1: Creation of Family Expenses Table

1. Go to All > In the filter search for Tables > click on New.
2. Enter the Details:
 Label : Family Expenses
 Name : Auto-Populated
 New menu name : Family Expenditure
3. Go to the Header and right click there >> click on Save.

servicenow All Favorites History Workspaces Table - New Record Search

Table New record

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Family Expenses Application Global

* Name u_family_expenses

Extends table

Create module ☒

Create mobile module ☒

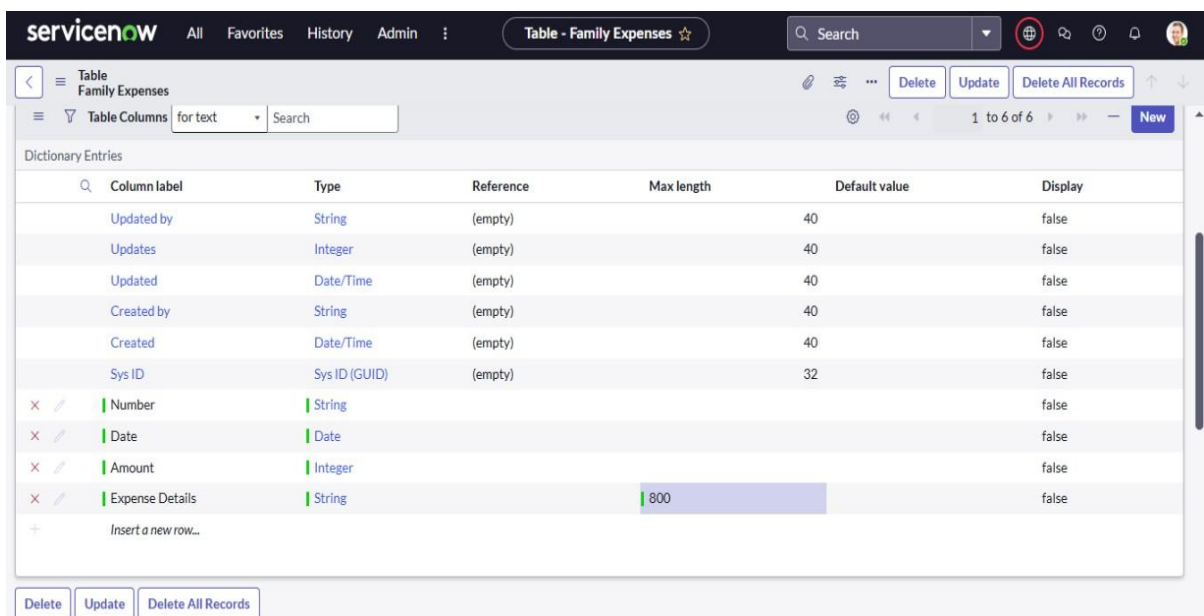
Add module to menu -- Create new --

New menu name Family Expenditure

Submit Cancel

Activity 2: Creation of Columns(Fields)

1. Near Columns Double click near insert a new row.
2. Give the details as:
 - i. Column label : Number
 - ii. Type : String
3. Double click on insert a new row again
4. Give the details as:
 - i. Column label : Date
 - ii. Type : Date
5. Double click on insert a new row again
6. Give the details as:
 - i. Column label : Amount
 - ii. Type : Integer
7. Double click on insert a new row again
8. Give the details as:
 - i. Column label : Expense Details
 - ii. Type : String
 - iii. Max length : 800\



Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Number	String				false
Date	Date				false
Amount	Integer				false
Expense Details	String		800		false
Insert a new row...					

9. Go to the Header and right click there>> click on Save.

Activity 3: Making Number Field an Auto-Number

1. Double click on the Number Field/Column.
2. Go down and double click on Advanced view
3. In Default Value:
 - i. Use dynamic default : check the box
 - ii. Dynamic default value : Get Next Padded Number
4. Click on Update.

Dictionary Entry
Number View: Advanced

Alters the behavior of a field or functionality that depends on the field. [More Info](#)

Attributes

Choice List Specification | Calculated Value | **Default Value**

The Default value specifies what value the field has when first displayed.

Use dynamic default ☒

Dynamic default value Get Next Padded Number

Delete Column Update

5. Go to All >> In the filter search for Number Maintenance >> select Number Maintenance
6. Click on New.
7. Enter the below Details:
 - i. Table : Family Expenses
 - ii. Prefix : MFE

servicenow All Favorites History Workspaces Number - New Record

Search

Number New record

* Table Family Expenses

Prefix MFE

* Number 1,000

Application Global

Number of digits 7

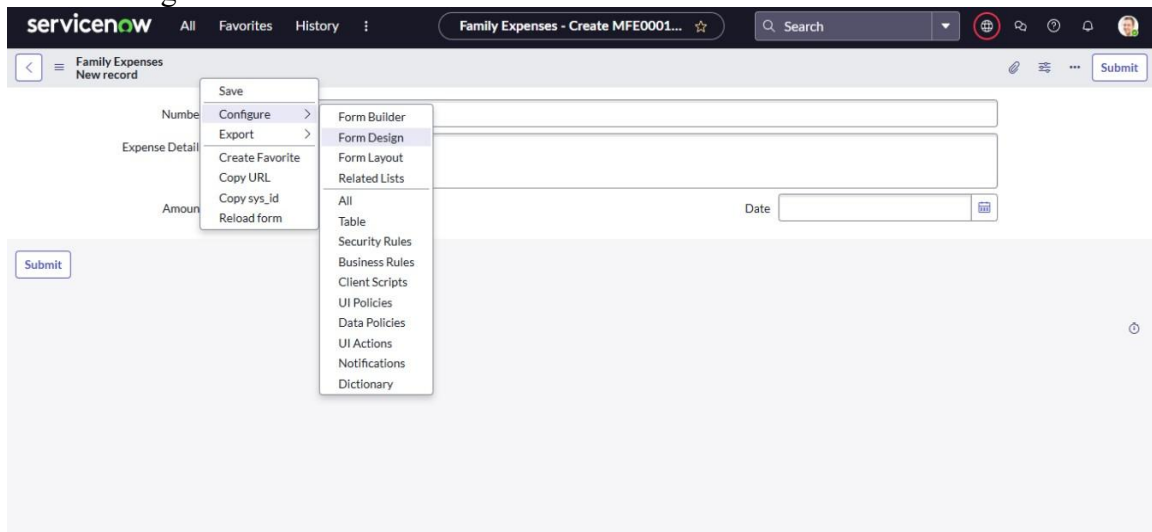
Submit

Related Links
[Show Counter](#)

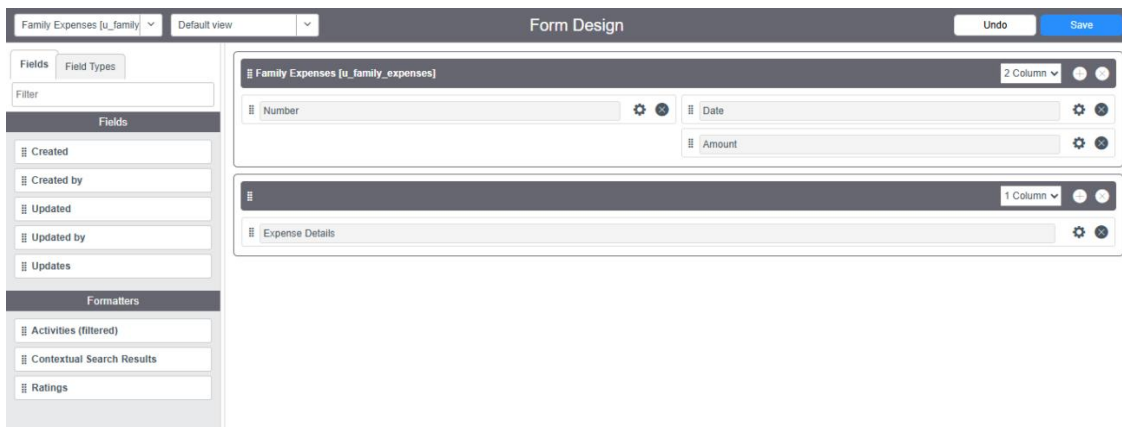
8. Click on Submit.

Activity 4: Configure the Form

1. Go to All >> In the filter search for Family Expenses >> Open Family Expenses
2. Click on New
3. Go to the Header and right click there>> click on Configure >> Select Form Design



4. Customize or Drag Drop the form as per your requirement.
5. Make Number Read-Only Field by clicking on the gear icon and checking Read-Only
6. Make Date, Amount Mandatory Field by clicking on the gear icon and checking Mandatory
7. Click on Save.



Milestone 5: Creation of Table (Daily Expenses)

Activity 1: Creation of Daily Expenses Table

1. Go to All > In the filter search for Tables > click on New.

The screenshot shows the ServiceNow 'Table - New Record' interface. At the top, there's a navigation bar with 'servicenow' logo and tabs for 'All', 'Favorites', 'History', and 'Workspaces'. A search bar and a 'Table - New Record' button are also present. Below the navigation bar, there's a breadcrumb trail 'Table > New record' and buttons for 'Submit' and 'Cancel'. A yellow banner states: 'ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).' Below this, a blue box explains: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'. The main form area contains several fields: 'Label' with the value 'Daily Expenses', 'Name' with the value 'u_daily_expenses', and 'Extends table' with a search icon. On the right side, there are checkboxes for 'Create module' and 'Create mobile module', both of which are checked. Below these, there's a dropdown menu for 'Add module to menu' with the value 'Family Expenditure'.

2. Enter the Details:
 - i. Label : Daily Expenses
 - ii. Name : Auto-Populated
 - iii. Add Module to menu : Family Expenditure
3. Go to the Header and right click there>> click on Save.

Activity 2: Creation of Columns(Fields)

1. Near Columns Double click near insert a new row.
Give the details as:
Column label : Number
Type : String
2. Double click on insert a new row again
Give the details as:
Column label : Date
Type : Date
3. Double click on insert a new row again
Give the details as:
Column label : Expense
Type : Integer
4. Double click on insert a new row again
Give the details as:
Column label : Family Member Name

Type : Reference
Max length : 800

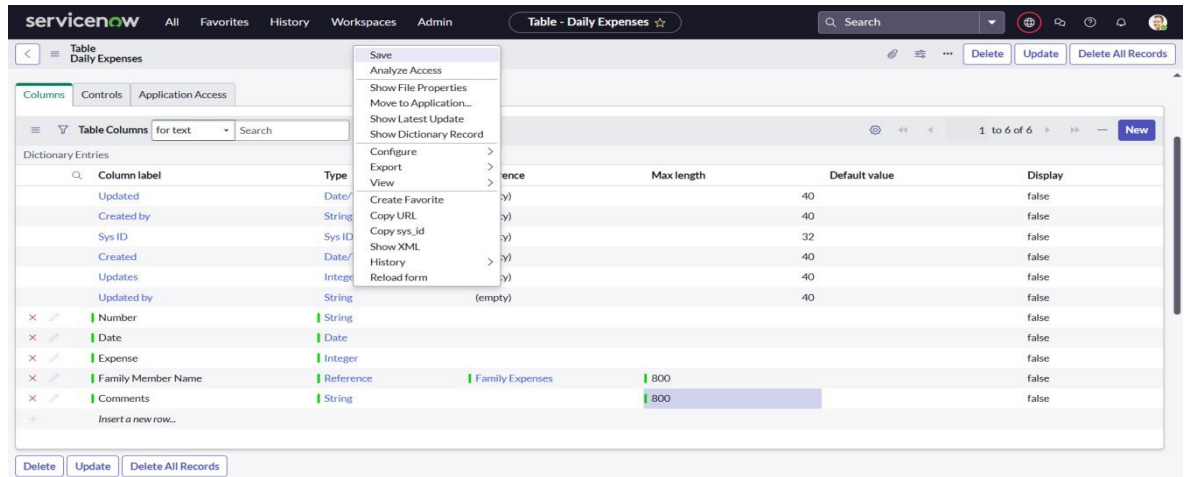
5. Double click on insert a new row again

Give the details as:

Column label : Comments

Type : String

Max length : 800



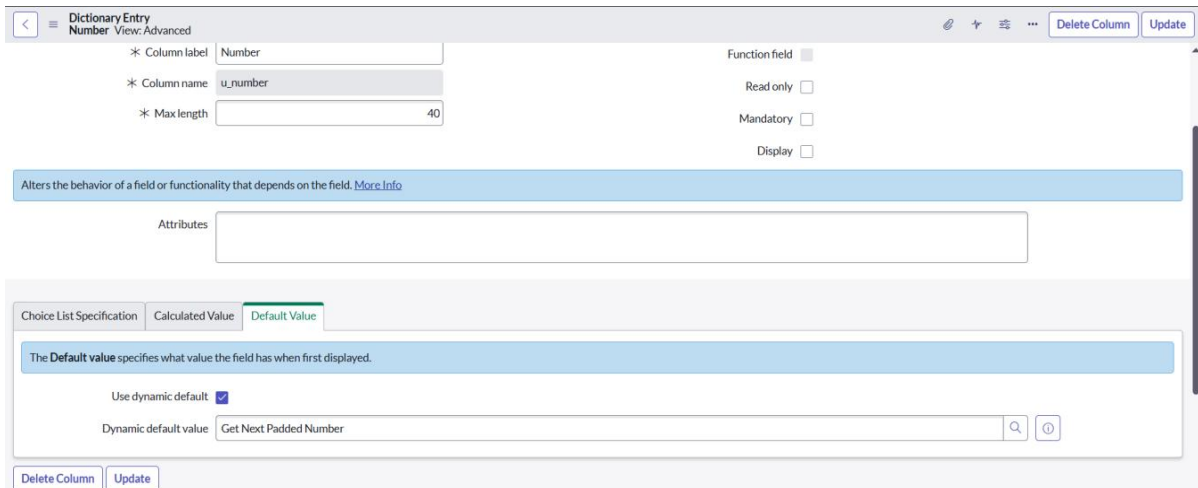
6. Go to the Header and right click there>> click on Save.

Activity 3: Making Number Field an Auto-Number

1. Double click on the Number Field/Column.
2. Go down and double click on Advanced view
3. In Default Value:

Use dynamic default : check the box

Dynamic default value : Get Next Padded Number



4. Click on Update.
5. Go to All >> In the filter search for Number Maintenance >> select Number Maintenance
6. Click on New.
7. Enter the below Details:

Table : Daily Expenses

Prefix : DFE

The screenshot shows the 'Number - New Record' form in ServiceNow. The form has the following fields:

- * Table: Daily Expenses
- Prefix: DFE
- * Number: 1,000
- Application: Global
- Number of digits: 7

At the bottom left, there is a 'Submit' button. At the bottom right, there is a 'Related Links' section with a link to 'Show Counter'.

8. Click on Submit

Activity 4 : Configure the Form

1. Go to All >> In the filter search for Daily Expenses >> Open Daily Expenses
2. Click on New
3. Go to the Header and right click there>> click on Configure >> Select Form Design

The screenshot shows the 'Daily Expenses - Create DFE0001001' form in ServiceNow. The form has the following fields:

- Number: DFE0001001
- Family Member Name:
- Comments:
- Date:
- Expense:

A context menu is open over the form, showing the following options:

- Save
- Configure >
- Export >
- Create Favorite
- Copy URL
- Copy sys_id
- Reload form
- Form Builder
- Form Design
- Form Layout
- Related Lists
- All
- Table
- Security Rules
- Business Rules
- Client Scripts
- UI Policies
- Data Policies
- UI Actions
- Notifications
- Dictionary

The 'Form Design' option is highlighted in the menu.

4. Customize or Drag Drop the form as per your requirement.
5. Make Number Read-Only Field by clicking on the gear icon and checking Read-Only
6. Make Date, Family Member Name Mandatory Field by clicking on the gear icon and checking Mandatory

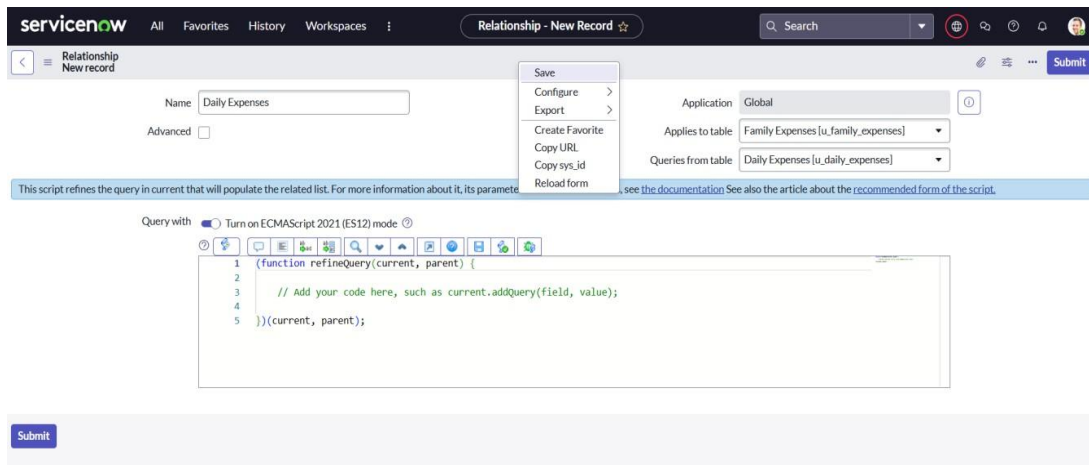
The screenshot shows a form editor for a table named 'Daily Expenses [u_daily_expenses]'. The form is organized into two columns. The first column contains two fields: 'Number' and 'Date'. The second column contains two fields: 'Family Member Name' and 'Expense'. Each field has a gear icon to its right, indicating that it can be configured. Below the main form is a 'Comments' section with a single text field and a gear icon. The interface includes a header bar with a title, a column selector (set to '2 Column'), and a footer bar with a title, a column selector (set to '1 Column'), and a 'Save' button.

7. Click on Save.

Milestone 6: Creation of Relationship

Activity : Creation of Relationship between Family Expenses and Daily Expenses tables.

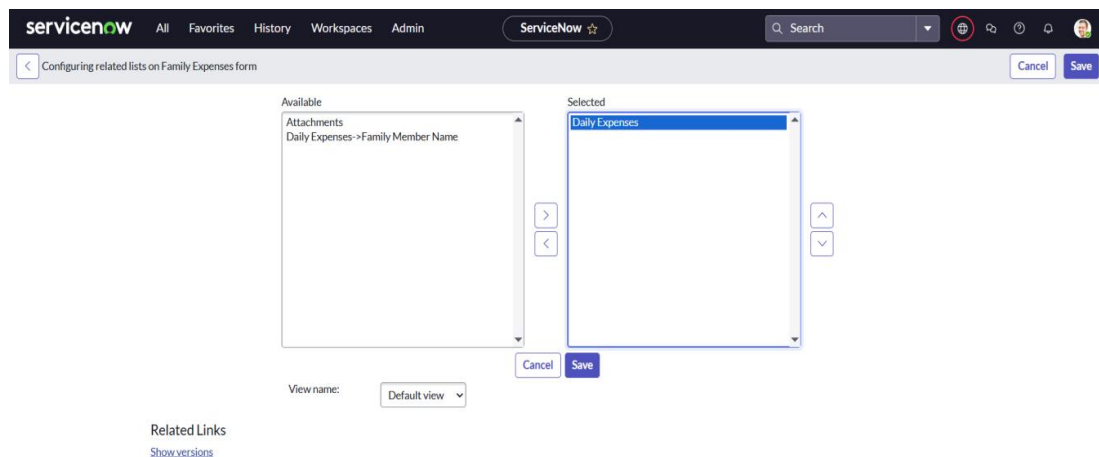
1. Go to All >> In the filter search for Relationships >> Open Relationships
2. Click on New.
3. Enter the details:
 - Name : Daily Expenses
 - Applies to table : Select Family Expenses
 - Daily Expenses : Select Daily Expenses
4. Click Save.



Milestone 7: Configuring Related List on Family Expenses

Activity : Configuring Related List on Family Expenses

1. Go to All >> In the filter search for Family Expenses >> Open Family Expenses
2. Click on New
3. Go to the Header and right click there>> click on Configure >> Select Related Lists
4. Add Daily Expenses to the Selected Area.



5. Click on Save

Milestone 8: Creation of Business Rules

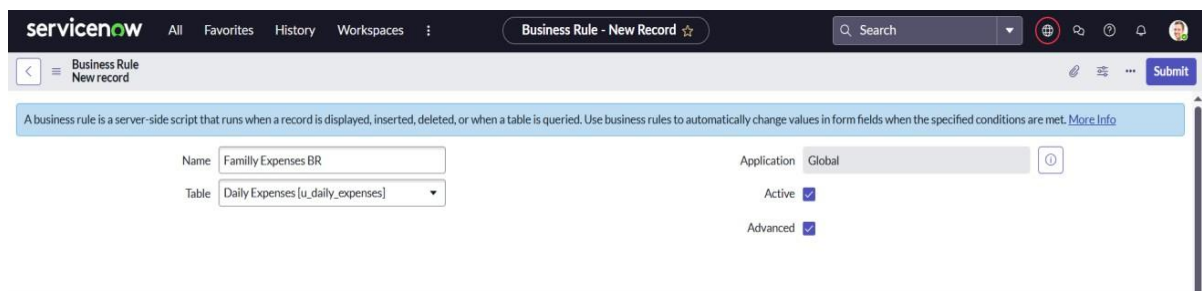
Activity : Creation of Business Rules

1. Go to All >> In the filter search for Business Rules.
2. Under System Definition Select Business Rules then click on New.
3. Enter the Details:

Name : Family Expenses BR

Table : Select Daily Expenses

Check Advanced



The screenshot shows the ServiceNow 'Business Rule - New Record' form. The header includes the ServiceNow logo, navigation tabs (All, Favorites, History, Workspaces), and a search bar. The form title is 'Business Rule - New Record'. Below the title, there is a description: 'A business rule is a server-side script that runs when a record is displayed, inserted, deleted, or when a table is queried. Use business rules to automatically change values in form fields when the specified conditions are met. [More Info](#)'. The form fields include: 'Name' (Family Expenses BR), 'Table' (Daily Expenses [u_daily_expenses]), 'Application' (Global), 'Active' (checked), and 'Advanced' (checked). A 'Submit' button is located at the top right.

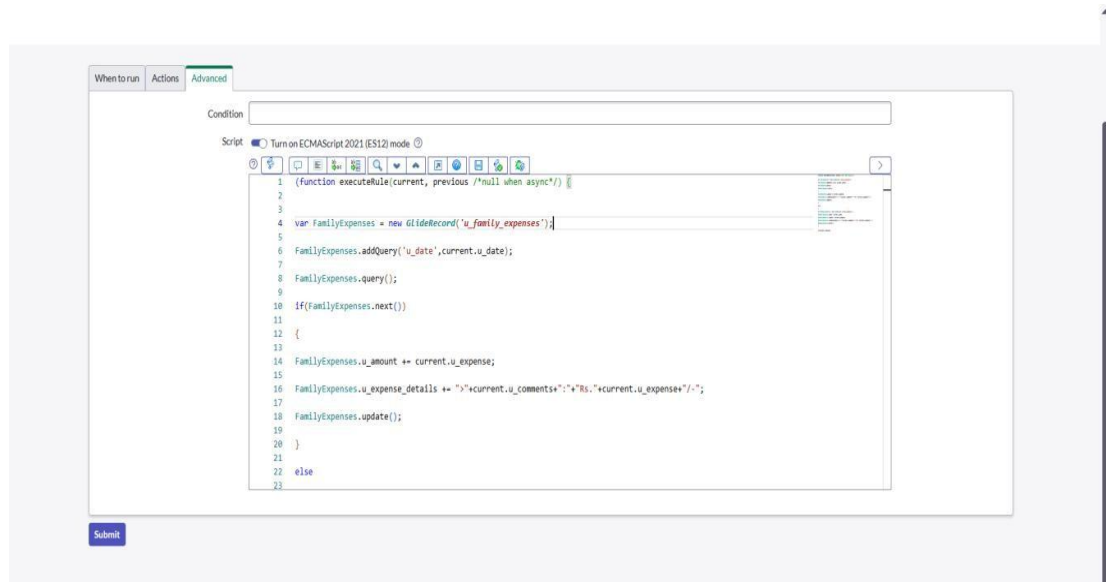
4. In when to run Check Insert and Update
5. In Advance(we write the code): Write the below

```
(function executeRule(current, previous /*null when async*/) {  
    var FamilyExpenses = new GlideRecord('u_family_expenses');  
    FamilyExpenses.addQuery('u_date',current.u_date);  
    FamilyExpenses.query();  
    if(FamilyExpenses.next())  
    {  
        FamilyExpenses.u_amount += current.u_expense;  
        FamilyExpenses.u_expense_details+=">"+current.u_comments+": "+"Rs."  
            +current.u_expense+"/";  
        FamilyExpenses.update()  
    }  
    else  
    {
```

```

var NewFamilyExpenses = new GlideRecord('u_family_expenses');
NewFamilyExpenses.u_date = current.u_date;
NewFamilyExpenses.u_amount = current.u_expense;
NewFamilyExpenses.u_expense_details+=">" +current.u_comments+": "+
    "Rs." +current.u_expense+"/-";
NewFamilyExpenses.insert();
}
})(current, previous);

```



6. Go to the Header and right click there>> click on Save.

Milestone 9 : Configure the Relationship

Activity : Configure the Relationship

1. Search for Relationships >> Open Relationships.
2. In that, open Daily Expenses Relationship.
3. For Applies to table : Select Family Expenses.
4. In Query with : write the below Query.

```

(function refineQuery(current, parent) {
// Add your code here, such as current.addQuery(field, value);
current.addQuery('u_date',parent.u_date)
current.query();
})(current, parent);

```

servicenow All Favorites History Workspaces Relationship - Daily Expenses

Name: Daily Expenses Application: Global

Advanced ☐ Applies to table: Family Expenses [u_family_expenses]

Queries from table: Daily Expenses [u_daily_expenses]

This script refines the query in current that will populate the related list. For more information about it, its parameters and control variables, see [the documentation](#). See also the article about the [recommended form of the script](#).

Query with ☐ Turn on ECMAScript 2021 (ES12) mode

```

1 (function refineQuery(current, parent) {
2
3
4 // Add your code here, such as current.addQuery(field, value);
5
6 current.addQuery('u_date',parent.u_date);
7
8 current.query();
9
10
11 })(current, parent);
12
13

```

Update Delete

5. Click on Update.

Conclusion :

The project “Calculating Family Expenses using ServiceNow” successfully demonstrates how technology can simplify and streamline the process of managing household finances. By leveraging the features of ServiceNow, the system enables effective expense tracking, categorization, and reporting, while maintaining scalability and security. Through the creation of customized tables, forms, relationships, and business rules, the project provides a user-friendly solution that empowers families to monitor their spending patterns and make informed financial decisions. Ultimately, this initiative not only promotes financial discipline but also highlights the potential of ServiceNow as a platform for developing practical, real-world applications beyond its traditional IT service management scope.